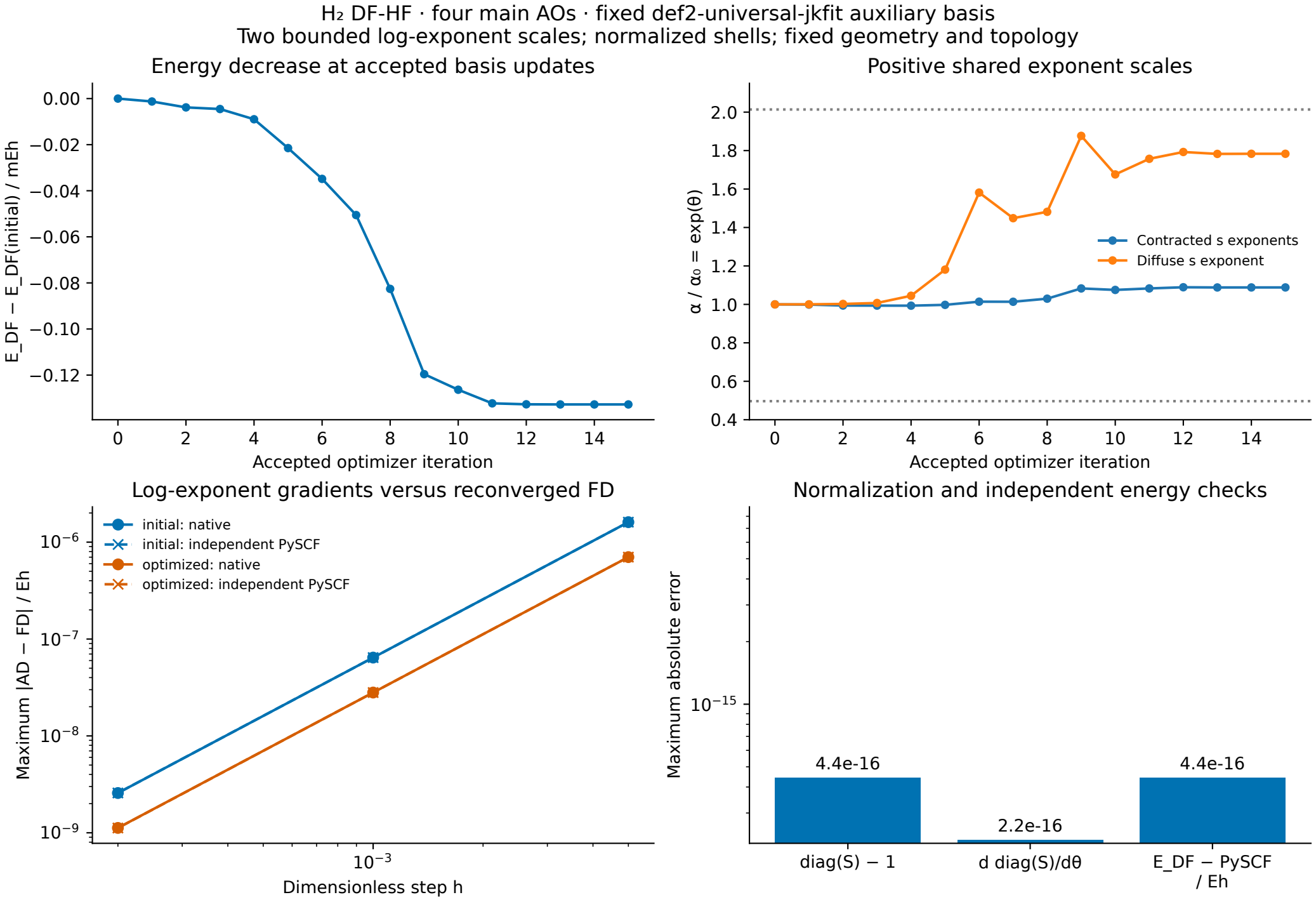
H₂ DF-HF · four main AOs · fixed def2-universal-jkfit auxiliary basis
Two bounded log-exponent scales; normalized shells; fixed geometry and topology
Energy decrease at accepted basis updates
E_DF − E_DF(initial) / mEh
Accepted optimizer iteration
Positive shared exponent scales
α / α₀ = exp(θ)
Contracted s exponents
Diffuse s exponent
Accepted optimizer iteration
Log-exponent gradients versus reconverged FD
initial: native
initial: independent PySCF
optimized: native
optimized: independent PySCF
Maximum |AD − FD| / Eh
Dimensionless step h
Normalization and independent energy checks
Maximum absolute error
4.4e-16
2.2e-16
4.4e-16
diag(S) − 1
d diag(S)/dθ
E_DF − PySCF / Eh